A fully specified defining quartic in 260 variables

14 November 2026

Abstract

The complete reference-point chart removes two rational coordinates from the fully specified 262-variable quartic. The new polynomial has 260 total variables, 232 quadratic residuals, degree exactly four, integer content one and 3704 monomials. Its rational input projection agrees with the predecessor for every fixed coefficient parameter. Consequently the same certified value $N = 9682$ and the same coefficient height $149^{38730} + 1$ are retained, under exactly the same arithmetic premises. No new prime-selection, height or pole-realization input is introduced. The same replacement also reduces the 226-variable existence family to 224 variables, although that is not the smallest existence family in this collection.

1 Statement

All counts include the input. The source is the fully specified 262-variable family of [1], at the retained main revision `fa09c9c6c97982c13265e7088d95cdb61b3e90cf`. Its quadratic-table SHA-256 is

`84c1130856dfda7d3a489f3077d7bc378cd8e594198dd8f3c03fe50463606a93`.

The coefficient parameter is $A = 149^{2N+1}$, never a quantified coordinate. Retain the exact elliptic group, normalized modular realization, pole-realization and fixed-data prime-pattern premises stated by that source.

Theorem 1.1. *There is a completely specified family $P_N \in \mathbb{Z}[X, Y_1, \ldots, Y_{259}]$, $N \geq 1$, with exact degree four, content one and 3704 nonzero coordinate monomials. For every fixed N, its rational input projection is equal to that of the source 262-variable polynomial. Its height is*

$$H(P_N) = 149^{4N+2} + 1 \qquad (N \geq 53).$$

Under the unchanged source arithmetic premises, every $N \geq 9682$ defines $\mathbb{Z}$ over $\mathbb{Q}$. In particular $N = 9682$ gives one fully specified polynomial of height $149^{38730} + 1$.

The source's effective threshold is preserved because the projection equivalence is for every fixed N, not merely for sufficiently large members with a new unspecified threshold. The algebraic equivalence itself is unconditional. The statement that either polynomial defines $\mathbb{Z}$ retains the source arithmetic inputs and their audit limitations.

2 The exact coordinate deletion

Work over $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$. The source reference block stores $f = 1/x(H)$, $S = y(H)/x(H)$, $g = 1/x(H - P)$ and $T = y(H - P)/x(H - P)$, with the complete three-field-quadratic fixed-difference chart. It also stores $B = f^2$ and $w = g^2$ for the local factors. The latter four rational output coordinates have no other definition required beyond these two squares.

In the literal source table the slots are

$$f = (x_{81}, x_{82}), \quad S = (x_{83}, x_{84}), \quad T = (x_{55}, x_{56}), \quad g = (x_{85}, x_{86}),$$

$$B = x_{132} + 2\rho x_{133}, \qquad w = x_{87} + 2\rho x_{88}.$$

All ordinary pairs in the first line use the basis $1, \rho$. The factor two in the second line is part of the source convention. Delete x_{85}, x_{86} and their two square-definition rows 23,24. Replace the six reduced reference rows 17–22 by the rational components of

$$Q_1(f, S, B, T) = 0, \qquad M(f, S, B)w + L(f, S, B, T) = 0, \qquad Q_3(f, S, B, T) = 0,$$

using exactly the corrected coefficient table in [2]. The already present rows 68,69 impose $B = f^2$ and are retained. Every other row is unchanged, apart from the final coordinate compaction. Inspection of all occurrences, performed by the checker, shows that deleted slots 85,86 occur only in the six reference rows and the two deleted square definitions.

Lemma 2.1. *The deleted coordinates are uniquely reconstructed from every rational solution of the new equations, and the reconstructed tuple solves the old equations. Conversely deleting these two coordinates from any old solution gives a new solution.*

Proof. The complete chart theorem [2, Theorem 1.1] applies over K, with no arithmetic hypothesis. It gives the actual point H, its translate, and the interpretations of f, S, B, w, T, including all exceptional leading-coefficient cases. In its notation,

$$G = \frac{((118 + 60\rho)S + 156 + 69\rho/2)T - (156 + 69\rho/2)S + 288 + 128\rho}{252 - 672\rho}, \qquad g = G^2/f.$$

The theorem proves $f \neq 0$, so this reconstructs a unique element of K, hence two rational coordinates. It also proves $w = g^2$ and all three original fixed-difference equations. The six old reduced rows are an invertible rational row transformation of those component equations, so they vanish. Their coefficient convention is retained in the source table. No other row consumes g, so all remaining old rows hold. Conversely every old reference point satisfies the four new field quadrics by the converse part of that same theorem; all unchanged rows agree literally. The input coordinate and N have not changed. □

Thus the new count is $262 - 2 = 260$ coordinates and $234 - 2 = 232$ rows. Their primitive integer sum of squares vanishes exactly when all rows vanish, proving projection equivalence.

3 Exact coefficient height and checks

The full producer and independent checker are in

`computation/quartic/compressed_reference_v27`.

The `family260` subdirectory includes the complete quadratic and expanded quartic tables. There are 3705 terms with powers of A kept separate, and 3704 ordinary coordinate monomials. All coordinate indices occur, degree is four, and the coefficient gcd is one.

Every coefficient profile is a fixed nonzero integer, a nonzero linear multiple of A, or the single profile $A^2 + 1$. The largest linear coefficient in absolute value is 44402. The full exact fixed maximum C_0 is recorded as an integer in the construction certificate. The checker proves by integer comparison

$$C_0 < 149^{214} + 1, \qquad 44402 < 149^{107}.$$

Consequently for $A \geq 149^{107}$ all constant and linear profiles are strictly smaller in absolute value than $A^2 + 1$. The latter is attained, proving the height formula for every $N \geq 53$. Positive

parameter values cannot cancel any of the stated profiles, so the monomial count is uniform. The unchanged effective source threshold 9682 is much larger, and is logically distinct from 53.

The independent checker imports no producer. It authenticates the source bytes, reconstructs all changed rows from the separately certified field-equation table, compares all unchanged rows, and expands every coefficient by a different term-pair enumeration. Changing a row or a quartic coefficient is rejected. The chart's own checker separately verifies all ideal identities and every exceptional case, including rejection of the omitted fS term. These computations are not a new audit of the source arithmetic premises and are not represented as non-author acceptance.

4 The corresponding 224-variable family

The 226-variable source has the same reference slots and square conventions. The identical deletion gives 224 variables, 198 rows, 3045 coordinate monomials and degree four. The source hash is

`ccc845018e4d89b9d19f797bbfeca10c17ec637fbde5cb5bc08ad3bba9da105b`.

The proof of Lemma 2.1 is unchanged. Its coefficient height is again $149^{4N+2}+1$ for $N \geq 53$, but its sufficient defining threshold remains ineffective, just as in that source. The complete tables and checks are in `family224`. This is an additional application of the exact chart equivalence, not a claim to improve the separate 204-variable existence construction.

References

[1] *A fully specified defining quartic in 262 variables*, retained project paper, `quartic262_20261110_v21.tex`.

[2] *A complete five-coordinate elliptic reference chart*, corrected project paper, `compressed_reference_20261114_v27.tex`.

[3] *A family of defining quartics in 226 variables*, retained project paper, `quartic226_20261110_v21.tex`.